

DoomOS v1.0.0 — Testing & Deployment Manual

This guide outlines the complete test and deployment procedures for **DoomOS (KDE Plasma 6 Edition)** across Virtual Machines (VMware Fusion / Workstation, UTM, QEMU) and Bare-Metal hardware.

Quick Launch Matrix

Environment	Host OS / Architecture	Command / Action	Acceleration Mode
VMware Fusion	macOS (Intel / x86_64)	<code>./test-vmware.sh`</code>	Direct 3D SVGA / Native
VMware Workstation	Linux / Windows (x86_64)	Copy <code>DoomOS.vmwarevm`</code>	Direct 3D SVGA / KVM / VT-x
UTM / QEMU	macOS (Apple Silicon arm64)	<code>./test-qemu.sh`</code>	TCG Multi-threaded Emulation
QEMU / KVM	Linux (x86_64)	<code>./test-qemu.sh`</code>	Hardware KVM (<code>`-enable-kvm`</code>)
Bare-Metal PC	Any x86_64 PC (AMD/Intel)	<code>./flash-usb.sh`</code>	Native GPU (AMD / Intel / NVIDIA)

Step 1: Virtual Machine Verification

Option A: Testing with VMware Fusion / Workstation

- Run `./test-vmware.sh`` to generate the pre-configured `DoomOS.vmwarevm`` bundle.
- The bundle contains:
 - Firmware**: UEFI Modern Boot (``firmware = "efi"``).
 - vCPU & RAM**: 4 Cores, 4096 MB RAM.

- **Storage**: High-performance NVMe virtual disk + SATA CD-ROM mounted to `output/doomos-plasma-x86\_64.iso`.
 - **Graphics**: 3D Acceleration enabled with 2048 MB VRAM (`mks.enable3d = "TRUE"`).
3. If on an Intel Mac or PC, start the VM:
- macOS: Double-click `DoomOS.vmwarevm` or run `/Applications/VMware\Fusion.app/Contents/Public/vmrun -T fusion start DoomOS.vmwarevm/DoomOS.vmx gui`.
 - Windows/Linux: Open `DoomOS.vmwarevm/DoomOS.vmx` in VMware Workstation.

Option B: Testing with UTM on Apple Silicon (M1/M2/M3/M4 Macs)

1. Run `/test-qemu.sh`.
2. When prompted, select **Y** to launch UTM.
3. In UTM:
 - Click the **+** (New Machine) button -> Select **Emulate** (Emulate slower, for different CPU architecture).
 - Select **Linux**.
 - Browse and select `doomos-builder/output/doomos-plasma-x86\_64.iso`.
 - Allocate **4096 MB RAM** and **4 CPU cores**.
 - Under Display, select **VirtIO-GPU**.
 - Click **Save** and hit **Play**.

Option C: Testing with Native Linux KVM

1. Run `/test-qemu.sh`.
2. The script auto-detects `/dev/kvm` and boots the ISO using hardware virtualization, 4 cores, 4GB RAM, and OVMF UEFI firmware.



Step 2: Live Session QA Checklist

Once booted into the Live Environment, verify each of the following subsystems:

1. Plymouth Boot & SDDM Autologin

- ☐ **Boot Animation:** Plymouth displays the DoomOS logo during kernel init without text corruption.

- **SDDM Autologin:** Boots directly into KDE Plasma 6 Wayland session as user `liveuser` without prompting for a password.
- **Wayland Verification:** Open Konsole and run:

```
```bash
```

```
echo $XDG_SESSION_TYPE
```

```
Expected output: wayland
```

```
```
```

2. High-DPI & Display Stack

- **Fractional Scaling:** Open *System Settings -> Display and Monitor*. Verify fractional scaling (125%, 150%) renders crisp text without Xwayland blur.
- **KWin Wayland Rules:** Confirm `/etc/xdg/kwinrc` has `AdaptiveSync=Always` and `AllowTearingAtFullscreen=true`.

3. Audio & Wayland Screen Sharing

- **PipeWire Audio Engine:** Verify PipeWire services are running:

```
```bash
```

```
systemctl --user status pipewire wireplumber
```

```
```
```

- **Virtual Meeting Loopback Sink:** Verify the loopback sink exists:

```
```bash
```

```
pactl list sinks short | grep -i loopback
```

```
Expected: DoomOS_Meeting_Share_Sink
```

```
```
```

- **Portal Verification:** Verify `xdg-desktop-portal-kde` is active for Wayland screen sharing:

```
```bash
```

```
systemctl --user status xdg-desktop-portal-kde
```

```
```
```

4. NVIDIA Guardian & Hardware Drivers

- Run the NVIDIA DKMS Guardian verification command:

```
```bash
```

```
doomos-nvidia-verify
```

```
```
```

- Confirm kernel modules exist under `/usr/lib/modules/$(uname -r)/`.`

5. Developer Tuning & Rootless Containers

- Verify sysctl values are applied:

```
```bash
```

```
sysctl fs.inotify.max_user_watches
```

```
Expected: 1048576
```

```
sysctl vm.max_map_count
```

```
Expected: 2147483642
```

```
```
```

- Confirm `/etc/subuid`` and `/etc/subgid`` are populated for rootless Podman/Docker.



Step 3: Installation & Btrfs Subvolume Verification

1. Launch Calamares

- Launch **Install DoomOS** from the desktop or application menu.
- Verify branding: Deep slate background with cyan/emerald highlights, product version ``DoomOS 1.0.0 Rolling``.

2. Disk Partitioning

- Select **Erase Disk** with **Btrfs** (default).
- Verify the installer allocates:
 - EFI System Partition (``/boot/efi``, FAT32).
 - Main Btrfs root filesystem.

3. Post-Install Subvolume Confirmation

After installation completes and the system reboots into the installed OS:

- Open Konsole and inspect Btrfs subvolumes:

```
```bash
```

```
sudo btrfs subvolume list /
```

```
```
```

Expected subvolumes:

- `@` mounted on `/`
- `@home` mounted on `/home`
- `@snapshots` mounted on `/.snapshots`
- `@var\_log` mounted on `/var/log`

4. Snapper Auto-Rollback Verification

- Verify Snapper root configuration:

```
```bash
```

```
sudo snapper -c root list
```

```
```
```

- Test automatic pacman pre/post snapshot:

```
```bash
```

```
sudo pacman -S --noconfirm fastfetch
```

```
sudo snapper -c root list
```

```
Confirm pre and post snapshot IDs were created by snap-pac
```

```
```
```

- Verify GRUB Btrfs boot integration:

```
```bash
```

```
systemctl status grub-btrfsd
```

```
```
```

5. Dual-Boot Windows RTC Sync

If installed alongside Windows in a dual-boot setup:

□ Verify local RTC status:

```
```bash
```

```
timedatectl | grep "RTC in local TZ"
```

```
Expected: RTC in local TZ: yes
```

```
```
```

Step 4: Bare-Metal USB Flashing

To flash the master ISO to a physical USB thumb drive:

1. Insert a USB flash drive (minimum 8 GB recommended).

2. Run the protected flashing utility:

```
```bash
```

```
./flash-usb.sh
```

```
```
```

3. The script will:

- Check the ISO's SHA256 checksum integrity.
- List only external/removable block devices.
- Block any attempt to write to internal or active system disks.
- Prompt for explicit `DOOM` typing confirmation.
- Write using high-speed block size (`4M`) and sync hardware buffers.

4. Plug the USB into your target PC, enter the UEFI boot menu (F11/F12/Del), and select ****UEFI: DoomOS****.